

Predicting Fuel Efficiency of an Automobile Through Regression

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Abstract

The fuel efficiency of a car is a critical concern in today's world of increasing environmental issues and rising fuel costs. Accurately predicting a car's miles per gallon (MPG) can benefit both the buyers and manufacturers in several ways. For car buyers, they can make informed decisions by knowing the expected expense for fuel and comparing it with alternate options. Predictions of MPG will assist manufacturers in designing more fuel-efficient cars which can help them gain a competitive advantage in the market. This paper is aimed to fit a straight-line model for predicting a car's MPG through various factors such as the number of cylinders, engine displacement, horsepower, weight of car, acceleration, model year, and the engine origin. Our results present a reliable linear regression model that can accurately predict a car's MPG with a margin of error of 0.551 miles per gallon and R_a^2 of 0.9055.

1 Introduction

A car's fuel efficiency is measured by miles per gallon (MPG), which indicates how many miles a car can travel on one gallon of fuel. Accurately predicting a car's MPG is not an easy task as numerous factors exist that influence its MPG value. Fortunately, there are some factors that have strong linear relationship with MPG that we may use them to build a straight-line model. The dataset used for this paper was taken from StatLib library at Carnegie Mellon University, and was also used in the 1983 American Statistical Association Exposition. With a sample of 398 distinct cars from the 1970 to 1982, we perform an exploratory data analysis to detect higher-order terms and multicollinearity. After resolving the issues of multicollinearity, we will hypothesize a complete second-order model and perform backward step regression to yield a more parsimonious model. Next, once a model is chosen, we will do a residual analysis to validate the random error assumptions of linear regression. Finally, we will discuss drawbacks of the project by presenting future improvements through cross-validation and sampling procedures.

2 Exploratory Data Analysis

Often, a simple exploratory data analysis can assist in the early stages of regression analysis by suggesting potential higher order terms and issues of multicollinearity. In this project, we were able to detect multicollinearity between independent variables by observing the pairplot between independent and dependent variables, which is shown in Figure 1. Engine's displacement had high correlation with horsepower, weight, and acceleration. As the quantity of air flowing through an engine increases, it translates into higher engine performance and greater weight, resulting in faster 0 to 60mph acceleration time. To identify and address multicollinearity, we used Variance Inflation Factor (VIF) to measure the amount an independent variable is influenced by other independent variables. The equation is shown in (1) where R_i^2 represents the coefficient of determination of the regression model of the i th independent variable on the remaining ones.

$$VIF_i = \frac{1}{1 - R_i^2} \quad (1)$$

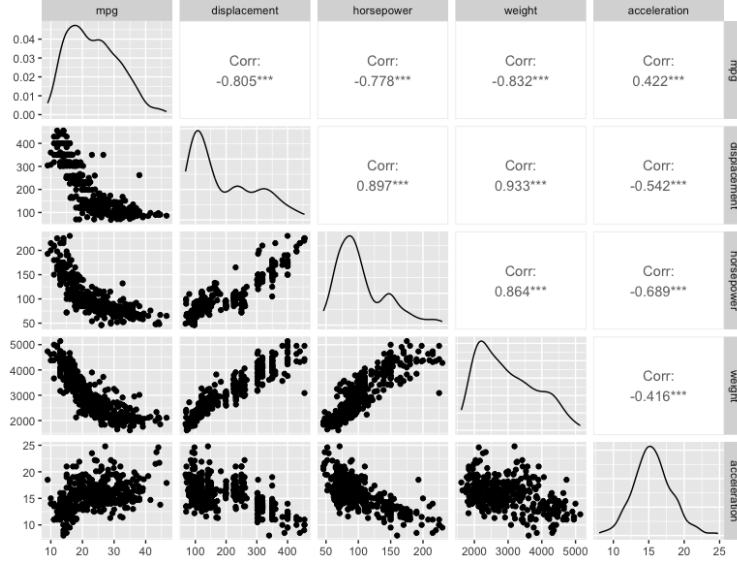


Figure 1: Pairplot of Independent and Dependent Variables.

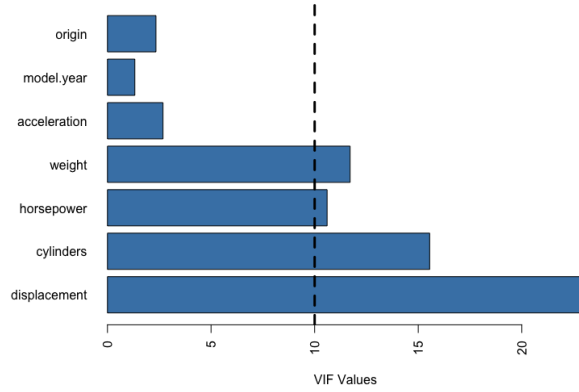


Figure 2: Barplot of VIF Values.

We set the threshold for VIF values at 10, and after referring to Figure 2, we decided to remove the variable engine displacement. By doing so, we were able to address the issue of multicollinearity.

Another discovery we made from the pairplot was the potential need to add second-order terms to our regression model. As seen in Figure 1, MPG had a curvilinear relationship with both horsepower and weight, suggesting the addition of quadratic terms to our model.

3 Model Hypothesis and Selection

Variable Notation

x_1 = horsepower (hp)

x_2 = weight (lb)

x_3 = acceleration (s)

x_4 = model year

$$x_5 = \begin{cases} 1, & \text{if 4 cylinders} \\ 0, & \text{if otherwise} \end{cases} \quad x_6 = \begin{cases} 1, & \text{if 5 cylinders} \\ 0, & \text{if otherwise} \end{cases} \quad x_7 = \begin{cases} 1, & \text{if 6 cylinders} \\ 0, & \text{if otherwise} \end{cases} \quad x_8 = \begin{cases} 1, & \text{if 8 cylinders} \\ 0, & \text{if otherwise} \end{cases}$$

$$x_9 = \begin{cases} 1, & \text{if Europe Origin} \\ 0, & \text{if otherwise} \end{cases} \quad x_{10} = \begin{cases} 1, & \text{if Asia Origin} \\ 0, & \text{if otherwise} \end{cases}$$

3.1 Initial Complete-Second Order Model

Since there are four quantitative variables and two qualitative variables, it is difficult to assume which variables may interact with each other. Therefore, a complete second-order model is an appropriate place to start.

$$\begin{aligned} E(y) = & \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4 + \hat{\beta}_5 x_1^2 + \\ & \hat{\beta}_6 x_2^2 + \hat{\beta}_7 x_3^2 + \hat{\beta}_8 x_4^2 + \hat{\beta}_9 x_1 x_2 + \hat{\beta}_{10} x_1 x_3 + \hat{\beta}_{11} x_1 x_4 + \\ & \hat{\beta}_{12} x_2 x_3 + \hat{\beta}_{13} x_2 x_4 + \hat{\beta}_{14} x_3 x_4 + \hat{\beta}_{15} x_5 + \hat{\beta}_{16} x_6 + \\ & \hat{\beta}_{17} x_7 + \hat{\beta}_{18} x_8 + \hat{\beta}_{19} x_9 + \hat{\beta}_{20} x_{10} + \hat{\beta}_{21} x_5 x_9 + \dots + \hat{\beta}_{224} x_3 x_5 x_{10} \end{aligned} \quad (2)$$

Equation (2) represents a complete second-order model that includes all possible combinations of k-way interaction terms between first and second-order quantitative variables and qualitative variables. However, a challenge with this equation is that, to estimate the 224 β coefficients, we would need at least 2230 observations according to the rule of thumb in linear regression.

3.2 Backward Step Regression

To address the issue of estimating a large number of β -coefficients, we will implement backward stepwise regression using R. By default, R uses the Akaike Information Criterion (AIC) to measure the relative quality of statistical models. However, we will instead use the Bayesian Information Criterion (BIC) to more severely penalize the model.

$$\text{AIC} = n \ln\left(\frac{SSE}{n}\right) + 2(k + 1) \quad (3)$$

$$\text{BIC} = n \ln\left(\frac{SSE}{n}\right) + (k + 1) \ln(n) \quad (4)$$

Note that, unlike AIC which uses a constant of 2 to penalize the model, BIC uses $\ln(n)$, where n is the number of observations. This results in a more parsimonious model selection criterion than AIC. After implementing backward stepwise regression using BIC in R, we obtained a model with 18 β -coefficients, which is sufficiently parsimonious to fit a straight-line model with 398 observations. The final selected model is presented in equation (5).

$$\begin{aligned}
E(y) = & \hat{\beta}_0 + \hat{\beta}_1x_1 + \hat{\beta}_2x_2 + \hat{\beta}_3x_3 + \hat{\beta}_4x_4 + \hat{\beta}_5x_2^2 + \\
& \hat{\beta}_6x_4^2 + \hat{\beta}_7x_9 + \hat{\beta}_8x_{10} + \hat{\beta}_9x_1x_4 + \hat{\beta}_{10}x_1x_9 + \hat{\beta}_{11}x_1x_{10} + \\
& \hat{\beta}_{12}x_3x_9 + \hat{\beta}_{13}x_3x_{10} + \hat{\beta}_{14}x_4x_9 + \hat{\beta}_{15}x_4x_{10} + \hat{\beta}_{16}x_1x_4x_9 + \hat{\beta}_{17}x_1x_4x_{10}
\end{aligned} \tag{5}$$

4 Residual Analysis: Validation of the Random Error Assumption

In order to build a reliable linear regression model, we must validate the assumptions for the random error component ϵ . There are four fundamental random error assumptions:

- ϵ is normally distributed
- $E(\epsilon) = 0$
- $\text{Var}(\epsilon)$ is constant
- Error terms are independent

The problem of our model is that it does not pass any graphical tests for random error assumption. We were able to detect model lack of fit by plotting residuals, $\hat{\epsilon}$, on the vertical axis against the predicted value, $\hat{y}$, on the horizontal axis. We observed a pattern of increasing variability as predicted value increased. Same was observed by plotting residual versus quantitative independent variables.

Our solution was to transform our dependent variable, MPG, by taking the natural logarithm of the values. By letting $y^* = \ln(y)$, we were able to resolve the problem of violating random error assumptions.

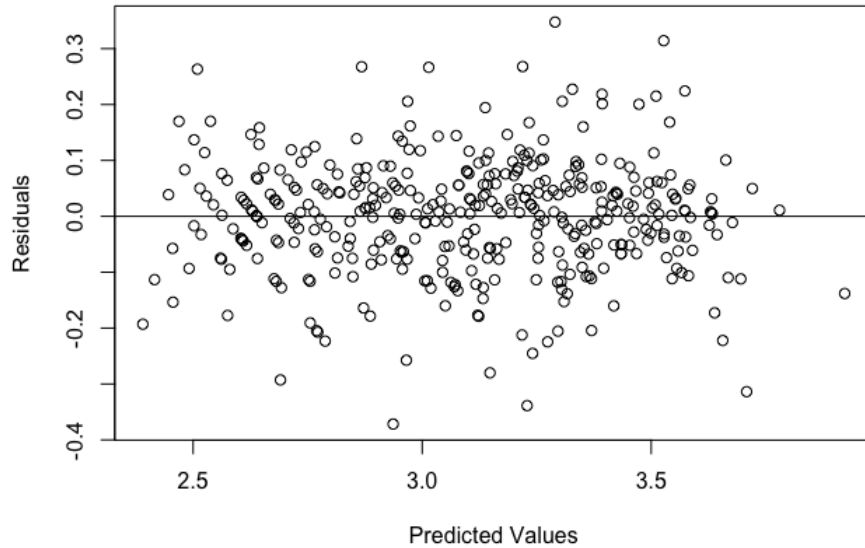


Figure 3: Residual Plot of Transformed Model versus Predicted Values

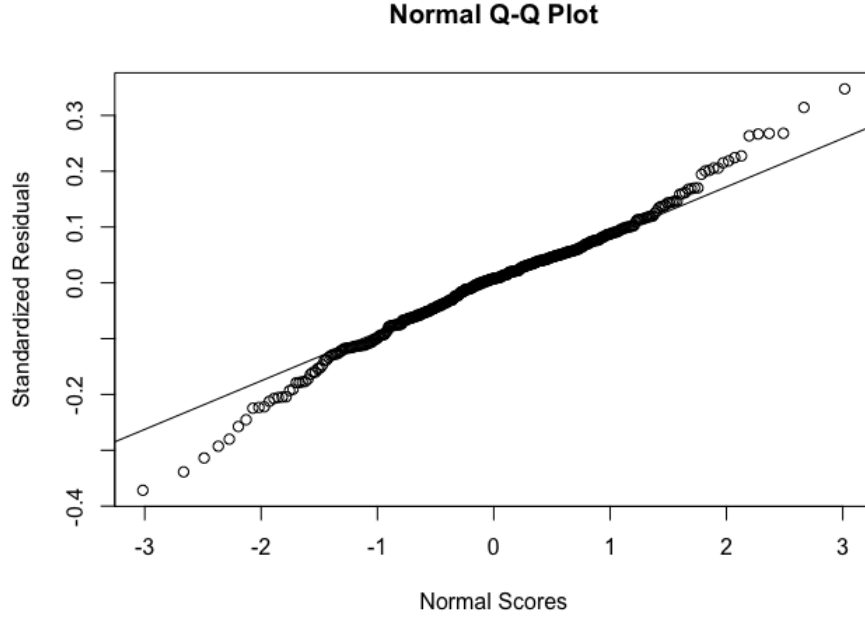


Figure 4: Normal Probability Plot of the Residuals

Figure 3 shows no pattern of the residuals as predicted values increase, which does not show signs of lack of fit. Normality assumption is met by observing the normal probability plot of the residuals shown in Figure 4. Since the straight line fits most of the points in the plot, we can safely assume that normality assumption is met. Although omitted, scatterplot of independent variables and the residuals showed no particular patterns to raise issues of violating random error assumptions.

5 Model Evaluation and Prediction

The final hypothesized model was fitted using R. The test of model utility shows that the model is statistically significant, as it resulted in an extremely small p-value. The model had a multiple R-squared of 0.906 and an adjusted R-squared of 0.9055, indicating that the regression model can explain about 91% of the sample variation. The residual standard error (s) was 0.1046, which, when raised to Euler's number e , results in a residual standard error of 1.11 MPG. In other words, about 95% of the sampled MPG values fall within $2s = 2.22$ miles per gallon of their predicted values. This level of accuracy, give or take 1.11 miles per gallon, is sufficiently good for both buyers and manufacturers to make informed decisions.

To put our model into use, we set a new car from 1970 with American engine, 8 cylinders, 165hp, 3693 pounds, 11.5 seconds of acceleration. The model's fitted value was 15.055 miles per gallon with lower and upper bound of the 95% confidence interval of 14.517, 15.613 respectively. The actual value of MPG was 15, which gives a margin of error 0.548 miles per gallon.

6 Conclusion/Discussion

In this paper, we fitted a linear regression model to a sample of 398 distinct cars to predict their MPG values using various factors, such as the number of cylinders, engine displacement, horsepower, weight of the car, acceleration, model year, and the engine origin. We hypothesized a complete second-order model

and then performed a backward step regression using Bayesian Information Criterion. We conducted a residual analysis using graphical methods to validate our random error assumptions. During the process, we transformed the dependent variable by taking the natural logarithm to normalize the residuals. Once the model was finalized, we fitted the linear regression model using R and evaluated our model.

One drawback of the model is that it cannot be used to make predictions for cars outside of the 1970-1982 time frame, as the dataset only contains information for those years. To generalize the model to today's automobiles, we would need to collect more recent data and fit the regression model again. Another concern is the relatively small number of observations, which could result in overfitting. To evaluate the external validity of the model, we could perform cross-validation if more data were available. Despite these limitations, the regression analysis demonstrates a strong ability to predict a car's MPG using readily available specifications.

References

- [1] Mendenhall, W., Sincich, T., Boudreau, N. S. (2003). A second course in statistics: regression analysis (Vol. 6). Upper Saddle River, NJ: Prentice Hall.